

16. A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

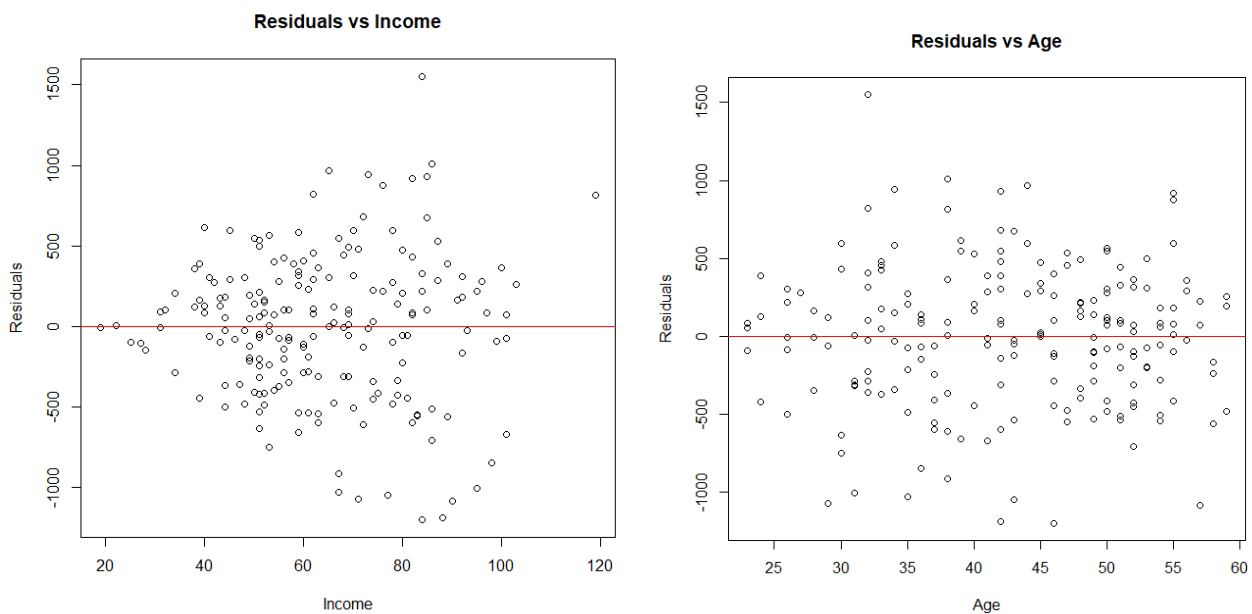
$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

a. Use the data file vacation to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

The 95% confidence interval for the coefficient on KIDS is [-135.3298, -28.3202]. Since the interval does not include zero, the effect of an additional child on miles traveled is statistically significant and negative, holding income and age constant.

b. Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?



1. Residuals vs. Income

The plot of residuals against Income reveals a clear fan-shaped pattern. As income increases, the dispersion of the residuals becomes larger. This provides strong visual evidence of heteroskedasticity, suggesting that the variance of the error term increases with income. In economic terms, higher-income individuals appear to exhibit greater variability in their behavior compared to lower-income individuals.

2. Residuals vs. Age

In the plot of residuals against Age, the pattern is less pronounced. The residuals appear more tightly clustered at younger ages and more dispersed in middle age ranges. While not a perfect funnel shape, this suggests possible mild heteroskedasticity with respect to age.

Overall, both plots are broadly consistent with the presence of heteroskedasticity, supporting the results of formal tests. This implies that OLS standard errors may be inconsistent, and therefore heteroskedasticity-robust standard errors or GLS-type methods would be more appropriate for inference.

c. Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

We test

$H_0: \sigma_{low}^2 = \sigma_{high}^2$, $H_1: \sigma_{low}^2 \neq \sigma_{high}^2$. To conduct the test, the sample is first sorted in ascending order according to income and then divided into two subsamples, consisting of the first 90 observations and the last 90 observations. Separate OLS regressions are estimated for each subsample, and the Goldfeld–Quandt test statistic is computed as the ratio of the estimated error variances.

The computed F-statistic is 3.1041, while the 5% critical value is 1.4286. Since the test statistic is greater than the critical value, we reject the null hypothesis. This provides evidence of heteroskedasticity, indicating that the variance of the error term differs between low-income and high-income households.

d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

The 95% confidence interval using heteroskedasticity-robust standard errors is [-139.3230, -24.3299]. Compared with part (a), the interval is slightly different due to the use of robust standard errors. However, both intervals remain entirely below zero, indicating that the effect of an additional child on miles traveled is significantly negative in both cases. Therefore, the qualitative conclusion remains unchanged.

e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

Based on the GLS estimation, the 95% confidence interval for the effect of one additional child on miles traveled is:

- Conventional GLS: [-119.8945, -33.7181]
- Robust GLS: [-121.4134, -32.1992]

Both intervals are very similar, although the robust GLS interval is slightly wider, reflecting the fact that it does not rely on the correct specification of the heteroskedasticity structure.

Compared with parts (a) and (d), the GLS-based intervals are generally more precise due to the efficiency gain from correctly modeling the error variance. However, all intervals (OLS, robust OLS, and GLS) lead to the same qualitative conclusion: an additional child significantly reduces miles traveled, as all confidence intervals remain entirely below zero.